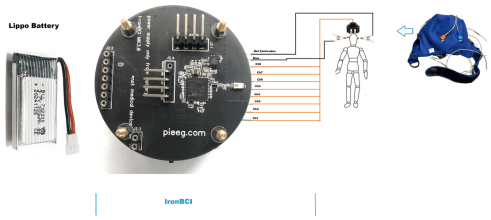
CONNECTOR, ELECTRODE & BATTERY GUIDE · SKU PIA24023P

Electrodes, Battery, Switches & Headers

How to connect 8 electrodes, the reference and the bias, the LiPo battery and its polarity, what the two switches and three LEDs do, and the SWD and I²C headers on the BLE board.



8	REF	BIAS	JST 2.0	2.54
Electrode channels	Ear clip · common 0 µV	Ear clip · noise rejection	LiPo connector	mm header pitch

01 ELECTRODE INPUTS

Input	Count	Electrode	Purpose
Channels 1–8	8	Dry or wet Ag/AgCl, in the cap	Voltage between each electrode and the reference
Reference	1	Ear clip on one earlobe	Common 0 µV point for all 8 channels
Bias	1	Ear clip on the other earlobe	Drives the body to cancel common-mode noise such as 50/60 Hz hum

02 ELECTRODE HEADER (ADC BOARD)

Square pad = pin 1 (E1). 2.54 mm pitch, single row.

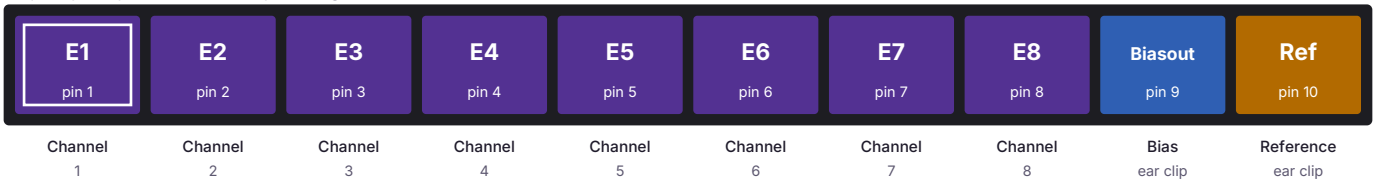
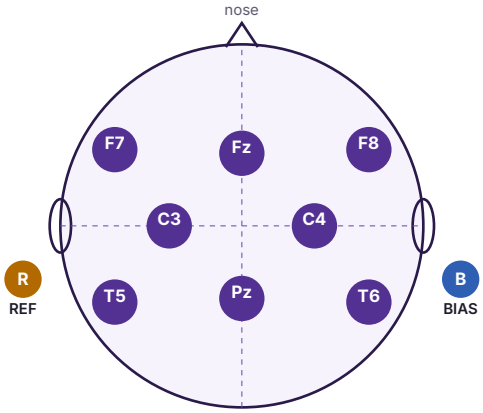


Figure 1 — Electrode header pin order. On the board the silkscreen reads Ref, Biasout, E8 ... E1 from the top.

> **NOTE Reference and bias clips matter most.**
> A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly before looking at anything else. Use 2.54 mm (0.1") female connectors.

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03 SUGGESTED MONTAGE (10-20 SYSTEM)



Position	Area	Good for
F7, F8	Front left / right, side	Eye movement, frontal activity
Fz	Front centre	Blinks, attention
C3, C4	Centre left / right	Motor activity (hands)
T5, T6	Back left / right, side	Temporal activity
Pz	Back centre, top	Alpha rhythm

Positions used for the dry-electrode tests in the IronBCI paper. Assign them to channels 1–8 in any order and note which went where — the software only knows channel numbers.

04 BATTERY

Type	Single-cell (1S) LiPo, 3.7 V nominal
Capacity	200 mAh recommended
Connector	2.0 mm JST, 2 pins
Location	Left edge, underside of the power board
Charging	Micro USB on the power board, 5 V supply



Battery on the underside of the stack.

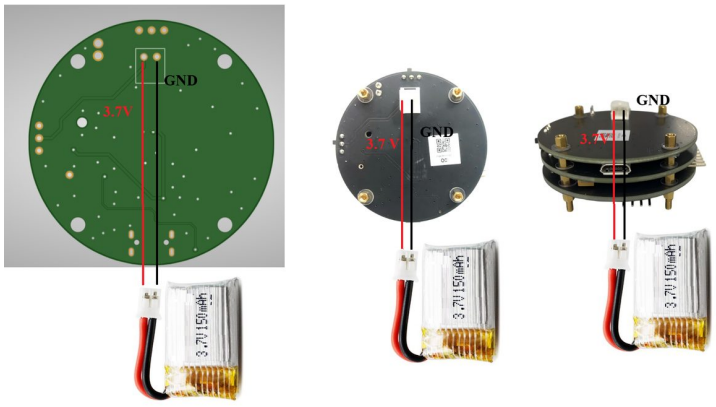


Figure 2 — Match the battery's black wire to the GND marking on the board.

WARNING Check polarity on every new battery.

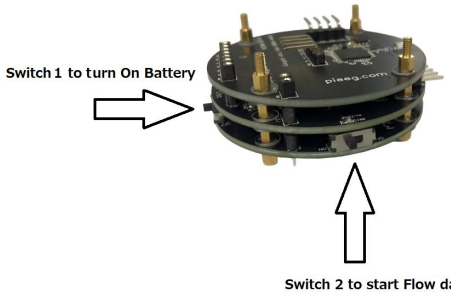
JST batteries are not all wired the same way. A reversed battery **will damage the board**. Match black to **GND** before plugging in.

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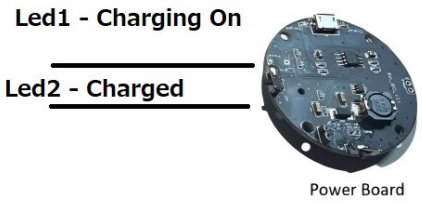
05 SWITCHES AND LEDS

Mode	Switch 1 (battery)	Switch 2 (data)
Recording	ON	ON
Charging	ON	OFF
Off / storage	OFF	OFF

LED	Meaning
LED 1 (power board)	Charging
LED 2 (power board)	Battery full
BLE LED	Bluetooth connection active



Switch 1 turns on the battery, Switch 2 starts the data flow.



LED 1 charging, LED 2 charged.

06 BLE BOARD HEADERS

SWD header J2.1 — programming

Pin	Signal	ST-Link V2
1	RST	RST
2	SWDIO	SWDIO
3	GND	GND
4	SWCLK (TCK)	SWCLK

No supply pin: power the device from its own battery (Switch 1 ON) while flashing.

I²C header J2.4 — expansion

Silk	Function
Reserv	Reserved, leave open
GND	Ground
3.2V	Supply for a small sensor (low current)
SCL	I²C clock
SDA	I²C data

Needs your own firmware; the standard firmware does not read I²C devices.

The 7-pin and 2-pin stacking headers between the boards are internal. Do not connect anything to them.

07 RELATED DOCUMENTS

IB8-DS-001 Product Datasheet	**IB8-MD-002** Mechanical & Assembly Drawing	**IB8-PCB-003** PCB & Manufacturing Specification
IB8-QSG-005 Quick Start & Safety Guide	**IB8-SW-006** Bluetooth Interface & Software Guide	

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